

VARFU OMU, Romania

WMO: 152800

Lat: 45.4460N

Lon: 25.4563E

Elev: 2506

StdP: 74.62

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.9	-19.8	-28.8	0.4	-13.2	-25.1	0.5	-13.9	34.3	-9.9	34.1	-10.2	12.2	320	1.082

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.7	14.6	10.3	13.1	9.4	11.8	8.7	11.5	13.3	10.3	12.0	9.4	11.0	3.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.7	10.9	12.3	9.6	10.1	11.1	8.7	9.5	10.1	40.4	13.4	37.2	12.1	34.7	11.0	15.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
33.5	24.4	20.1	DB	-25.0	17.4	2.3	1.9	-26.6	18.7	-28.0	19.8	-29.2	20.9	-30.9	22.3	(4)
			WB	-25.1	13.5	2.3	1.3	-26.7	14.4	-28.1	15.2	-29.3	15.9	-31.0	16.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.6	-10.0	-9.7	-8.1	-3.5	1.3	5.0	7.0	7.4	3.3	-0.1	-4.0	-8.2	(6)
(7)		DBStd	7.53	5.03	4.84	4.51	3.82	3.48	3.51	3.18	3.11	3.88	4.79	5.20	5.01	(7)
(8)		HDD10.0	4252	620	551	560	404	269	152	103	92	203	314	419	563	(8)
(9)		HDD18.3	7267	878	784	818	654	528	399	352	338	451	572	669	822	(9)
(10)		CDD10.0	27	0	0	0	0	0	3	9	12	2	1	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	8.1	10.1	10.0	9.6	7.9	7.5	6.5	6.1	5.4	7.1	7.5	9.6	9.6	(14)
(15)	Precipitation	PrecAvg	885	46	42	52	70	105	119	113	96	74	63	53	52	(15)
(16)		PrecMax	1240	125	83	93	144	195	217	197	207	147	136	107	122	(16)
(17)		PrecMin	575	8	7	13	35	35	52	12	31	20	4	4	8	(17)
(18)		PrecStd	166	27	19	22	28	41	45	48	46	35	32	27	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	1.6	2.2	6.6	10.7	14.9	16.5	16.6	14.6	11.8	7.8	4.5	(19)
(20)		0.4%	MCWB	-2.5	-1.5	-0.6	2.2	7.4	10.0	11.1	11.7	8.8	6.0	2.3	-1.0	(20)
(21)		2%	DB	-0.9	-0.8	0.4	4.4	8.5	12.7	14.4	14.7	11.6	9.4	5.7	0.9	(21)
(22)		2%	MCWB	-4.0	-2.8	-1.5	1.8	5.7	9.2	10.1	10.8	7.1	3.5	1.1	-2.1	(22)
(23)		5%	DB	-2.5	-2.5	-1.0	2.6	7.1	11.1	12.8	13.1	9.6	7.5	4.0	-1.0	(23)
(24)		5%	MCWB	-4.1	-3.9	-2.2	0.7	4.8	8.3	9.4	9.6	5.9	2.8	-0.4	-3.1	(24)
(25)		10%	DB	-3.8	-3.6	-2.3	1.1	5.8	9.7	11.4	11.6	8.1	5.9	2.3	-2.2	(25)
(26)		10%	MCWB	-4.9	-4.8	-3.1	-0.1	4.0	7.5	8.9	8.7	5.3	2.3	-0.9	-3.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.2	-0.6	0.5	4.0	8.2	12.0	12.9	13.5	10.1	8.5	4.7	0.7	(27)
(28)		0.4%	MCDWB	-0.5	0.6	1.3	5.6	9.6	13.5	14.8	15.5	12.9	10.2	6.1	2.0	(28)
(29)		2%	WB	-2.7	-2.4	-0.9	2.2	6.6	10.2	11.3	11.7	8.2	5.7	2.3	-1.3	(29)
(30)		2%	MCDWB	-2.0	-1.5	-0.3	3.6	7.9	11.9	13.0	13.5	10.2	7.2	3.6	-0.1	(30)
(31)		5%	WB	-3.8	-3.5	-2.0	1.1	5.4	9.0	10.2	10.4	7.0	4.3	1.2	-2.3	(31)
(32)		5%	MCDWB	-3.1	-2.8	-1.3	2.2	6.4	10.4	11.8	12.1	8.7	5.8	2.4	-1.5	(32)
(33)		10%	WB	-4.8	-4.6	-3.1	0.1	4.4	7.9	9.3	9.4	6.1	3.2	0.1	-3.3	(33)
(34)		10%	MCDWB	-4.0	-3.8	-2.4	0.8	5.5	9.2	10.8	10.9	7.4	4.7	1.4	-2.4	(34)
(35)	Mean Daily Temperature Range		MDBR	4.7	4.9	5.2	4.7	4.3	4.5	4.6	4.7	4.5	4.9	4.7	4.7	(35)
(36)		5% DB	MCDBR	5.4	5.6	6.1	5.8	5.8	6.3	6.4	6.3	6.1	6.0	5.8	5.2	(36)
(37)			MCWBR	4.7	5.0	5.5	4.9	5.4	5.8	5.9	6.2	5.9	5.5	4.4	4.4	(37)
(38)			MCDBR	5.1	5.2	5.5	5.0	5.3	5.7	5.8	5.8	5.4	5.4	4.5	4.5	(38)
(39)		5% WB	MCWBR	4.8	5.0	5.4	4.6	5.2	5.6	5.8	6.0	5.5	5.5	4.3	4.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.223	0.240	0.266	0.303	0.295	0.297	0.306	0.313	0.287	0.264	0.247	0.225	(40)	
(41)		taud	2.467	2.442	2.474	2.449	2.500	2.504	2.470	2.443	2.519	2.588	2.590	2.513	(41)	
(42)		Ebn at Noon	929	967	978	960	973	968	955	937	942	928	893	892	(42)	
(43)		Edh at Noon	65	82	93	105	103	104	106	105	88	71	59	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.40	2.22	3.29	4.29	5.22	5.65	5.75	5.20	3.81	2.64	1.54	1.15	(44)	
(45)		RadStd	0.14	0.28	0.34	0.47	0.51	0.51	0.49	0.43	0.37	0.34	0.16	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.52	N/A	N/A	N/A	N/A	N/A	-183	-194	N/A	N/A	(46)
(47)	Regional (69 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	-121	-165	+65	+17	(47)

Nomenclature: See separate page